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Course on Advanced Computer Architectures

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EX1	(6 points)	
EX2	(5 points)	
EX3	(5 points)	
Q1	(5 points)	
Q2	(5 points)	
SIX QUESTIONS	(6 points)	
TOTAL	(32 points)	

EXERCISE 1 – SCOREBOARD (6 points)

PART A: Please consider the program in the table be executed on a CPU with dynamic scheduling based on **SCOREBOARD BASIC SCHEME** with the following assumption:

- 2 LOAD/STORE Units (LDU1, LDU2) with latency 3 cycles
- 1 FP Unit (FPU1) with latency 3 cycles
- 1 ALU/BR Unit ALU1 with latency 1 cycle

INSTRUCTION	ISS UE	READ OPs	EXEC COMPL.	WRITE BACK	Hazards Type	UNIT
I0:ld \$f2,VECTB(\$r6)						
I1:faddi \$f3,\$f2,K						
I2:st \$f3,VECTA(\$r7)						
I3:ld \$f3,VECTC(\$r6)						
I4:st \$f3,VECTC(\$r7)						
I5:addi \$r6,\$r6,4						
I6:addi \$r7,\$r7,4						

Express the formula and calculate the CPI:

CPI = _____

PART B: Please consider the introduction of the following **Optimizations**:

- Check for WAW postponed to WRITE BACK phase
- Forwarding

INSTRUCTION	ISS UE	READ OPs	EXEC COMPL.	WRITE BACK	Hazards Type	UNIT
I0:ld \$f2,VECTB(\$r6)						
I1:faddi \$f3,\$f2,K						
I2:st \$f3,VECTA(\$r7)						
I3:ld \$f3,VECTC(\$r6)						
I4:st \$f3,VECTC(\$r7)						
I5:addi \$r6,\$r6,4						
I6:addi \$r7,\$r7,4						

Calculate the CPI: CPI = _____

Calculate the Speedup with respect to the SCOREBOARD BASIC SCHEME used in PARTA:

SpeedUP = _____

EXERCISE 2: VLIW (5 points)

In this problem, you will port code to a simple **3-issue VLIW machine**, and schedule it to improve performance. Details about the 3-issue VLIW machine with 3 fully pipelined functional units:

- Integer ALU with **1 cycle** latency to next Integer/FP and **2 cycle** latency to next Branch
- Memory Unit with **3 cycle** latency
- Floating Point Unit with **3 cycle** latency (each FPU can complete one add or one multiply per clock cycle)
- Branch completed with **1 cycle delay slot** (branch solved in ID stage)
- No interlocks

C Code:

```
for (j = 0, i = 1; i < N; i++,  
j++) {  
    vectA[i] = vectB[j] + K;  
    vectC[i] = vectC[j];  
}
```

Assembly Code:

```
for: ld f1, vectB(r6)  
     faddi f3, f2, K  
     st f3, vectA(r7)  
     ld f3, vectC(r6)  
     st f3, vectC(r7)  
     addi r6, r6, 4  
     addi r7, r7, 4  
     blt r7, r8, for
```

Considering **one iteration** of the loop, **schedule** the assembly code for the 3-issue VLIW machine in the following table by using the **list-based scheduling**, but neither using software pipelining nor loop unrolling **nor modifying loop indexes**. Please do not need to write in NOPs (can leave blank).

	Integer ALU	Memory Unit	FPU
C0			
C1			
C2			
C3			
C4			
C5			
C6			
C7			
C8			
C9			
C10			
C11			
C12			
C13			
C14			

How long is the Critical Path? _____

What performance did you achieve in FP ops per cycle? _____

What performance did you achieve in cycles per loop iteration? _____

What code efficiency did you achieve? _____

What CPI did you achieve? _____

EXERCISE 3: CACHE COHERENCY (5 points)

1. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the **MESI protocol** is used, with **write-back** caches, **write-allocate**, and **write-invalidate** of other caches.

Cycle	After Operation	P0 cache block state	P1 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P0: read block 1	Exclusive (1)	Invalid	Yes	Yes
1	P1: read block 0				
2	P0: write block 1				
3	P0: write block 0				
4	P1: read block 0				
5	P1: write block 0				
6	P0: read block 1				
7	P1: read block 1				
8	P0: write block 1				
9	P1: write block 1				
10	P0: read block 0				
11	P1: write block 1				
12	P1: read block 1				
13	P0: read block 1				
14	P1: write block 1				

QUESTION 1: MEMORY HIERARCHY (5 points)

Considering the data policies in a caching system with a small L1-cache and a larger L2-cache, in your opinion which can be the rationales behind them?

For each answer please indicate if they are **TRUE** or **FALSE** and explain **WHY**

Answer 1: With exclusive policy the total capacity of the cache is larger than using inclusive policy. **T F**

Answer 2: With inclusive policy the miss penalty is smaller w.r.t exclusive policy **T F**

Answer 3: The global miss rate represents the fraction of memory accesses going from the processor to the main memory **T F**

Answer 4: The introduction of a larger L2-cache reduces both the effective miss penalty and the average memory access time **T F**

- Present briefly the **explicit** register renaming mechanism, explaining how it works
- Explain the differences with respect to an **implicit** register renaming approach such as Tomasulo
- Explain how the two approaches can guarantee precise interrupts

[illegible]

SIX QUESTIONS: (6 points)

Answer with **True** or **False** to the following questions:

Q1) Fine-grained multithreading aims at speeding up the execution time of each individual thread.

Answer: T F

Q2) In the speculative Tomasulo architecture, exceptions are taken when the instruction that generated them reaches the head of the ROB.

Answer: T F

Mark the **True** answers to the following questions (some questions might have **multiple True answers**):

Q3) Which of the following sentences represent the main Tomasulo drawbacks?

Answer 1: It requires a large amount of hardware.

Answer 2: The performance might be limited using a single Common Data Bus.

Answer 3: Reservation Stations limit the amount of functional units that can be used.

Answer 4: It doesn't have a way to handle RAW hazards

Q4) Consider a five stage MIPS pipeline with precise exceptions, which statements concerning exception handling are correct?

Answer 1: Since the instructions are issued in-order the exceptions occur in-order.

Answer 2: Exceptions can be handled as soon as they occur.

Answer 3: Exceptions are handled in the order in which they would occur in an unpipelined processor.

Answer 4: It's possible for instructions executing after a (not yet handled) exception to change the state of the processor.

Q5) What are the issues that led to the necessity of multiple cores?

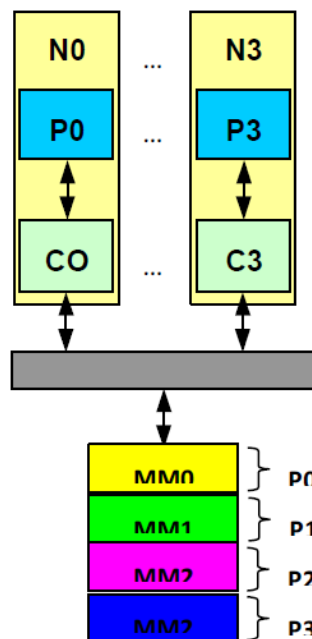
Answer 1: Deep pipelines cause heat dissipation problems.

Answer 2: Area occupation lack in modern chips.

Answer 3: One core can't execute correctly an instruction trace with dependencies.

Answer 4: It is difficult to push the limits of the improvement of a single core.

Q6) *The architecture in the following picture represents:*



Answer 1: Centralized Memory; Single Logically Shared Address Space

Answer 2: Centralized Memory; Message Passing Architectures

Answer 3: Distributed Memory; Single Logically Shared Address Space

Answer 4: Distributed Memory; Message Passing Architectures